sterilized containers to prevent recontamination. The degree of heat and the length of time of heating vary with the type of food and the kinds of micro organisms. Large quantities of food are canned for preservation. In developed countries, canned foods form a major part of the diet of the people. *Items* often canned are *meats and meat products, fruits and vegetables, fish products, soups, etc.*



The process of canning involves the following steps:

- *Receiving, cleaning, grading and inspecting of raw commodity.*
- *Blanching to inactivate enzymes.*
- *Placing in the container with added brine or syrup and deaeration of the product.*
- *The next process is exhausting. Exhausting is done to expel the air and gas from the can so that its internal pressure, after heating and cooling, is the same as the atmospheric pressure.*
- *After exhausting, the filled cans are permanently sealed mechanically.*
- *The sealed containers are subjected to high temperatures, to destroy the most heat resistant organisms.*
- *After this, the cans are cooled by water in a cooling canal to about 38°C, before storage.*
- *The final step is casing and storing the cans.*



▲ Fig. 4 Home scale canning

4.2.3 Preservation by Dehydration

Dehydration is *the extraction of moisture* from food products like fruits, vegetables, herbs and meat. It inhibits the growth of microorganisms and *imparts a long storage life*. This is a modern development of smoking and drying. Some *changes* that occur during the process of dehydration are:

- *Chemical changes*
- *Browning and flavour changes*
- *Denaturation of proteins*
- *Concentration on the surface of the food (case hardening)*

Dehydration can be done by *drying and salting*. Evaporation is quickened with the addition of moderate heat which is sometimes provided *by natural sunlight*. The ultraviolet rays from the sun serve to *kill microbes*. Modern methods of dehydration use circulating air that is heated just enough to promote dehydration without cooking the food. Food preservation by drying is one of the oldest methods used by human beings. Drying is one of the methods used for dehydration.

4.2.3.1 Drying

Drying is the method nature resorts to preserve foods. Natural drying was adopted by early man to dry fruits, fish and meat by exposing them to the sun.

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